Credit Card Fraud Detection

End-to-End Machine Learning System — Technical Report

To: Head of Risk Analytics

From: Data Science — Risk & Fraud

Date: 23 April 2026

Dataset: IEEE-CIS Fraud Detection (Vesta Corporation, 2019)

Status: Internal Review Draft

Key Findings at a Glance

Metric	Value	Interpretation
Best Model	XGBoost	Highest PR-AUC on validation set
PR-AUC	0.824	Strong minority-class discrimination
Recall	78.7%	Fraud transactions correctly flagged
Precision	69.7%	Flagged transactions that are real fraud
F2 Score	0.767	Primary metric (recall-weighted)
Operating Threshold	0.446	F2-optimal classification boundary

1. Executive Summary

This report presents the findings of a machine learning fraud detection system trained on the IEEE-CIS transaction dataset (590,540 transactions, December 2017 to May 2018, sponsored by Vesta Corporation). The objective was to build a production-ready classifier that identifies fraudulent payment transactions with high recall — minimising the cost of missed fraud — while maintaining sufficient precision to keep analyst review workload manageable.

The winning model is **XGBoost**, tuned with Optuna over 50 trials. On the held-out validation set (118,108 transactions, 4,133 fraud), it achieves a **PR-AUC of 0.824**, catching **78.7% of all fraud** at a precision of **69.7%**. This represents a **169% improvement** in PR-AUC over the Logistic Regression baseline (0.306), demonstrating the value of gradient boosting on high-dimensional behavioural transaction features.

The system operates at two thresholds: an F2-optimal threshold (0.446) for the analyst review queue, and a high-precision threshold (0.990) for automatic hard-block decisions. All predictions are accompanied by SHAP-based feature importance explanations, enabling analysts to understand and audit every flagged transaction.

Primary Recommendation

Deploy XGBoost at the F2-optimal threshold as the primary fraud scoring engine. Route high-confidence predictions (above the hard-block threshold) to automatic decline. Route mid-confidence predictions to the analyst review queue with SHAP explanations attached. Monitor PR-AUC weekly and retrain quarterly as fraud patterns evolve.

2. Methodology

2.1 Dataset

The IEEE-CIS Fraud Detection dataset contains 590,540 real payment transactions from Vesta Corporation's payment network. Each transaction has 394 raw features across two tables: transaction attributes (amount, product code, card metadata, Vesta's proprietary V-features capturing behavioural signals) and identity attributes (device type, browser, network fingerprints). The two tables are joined on TransactionID via left join, preserving all 590,540 transactions regardless of whether identity data is available (only 24.4% of transactions have identity records — the missingness is itself a signal).

The dataset exhibits a fraud rate of 3.50% (20,663 fraud / 569,877 legitimate), a 27.6:1 class imbalance. After dropping columns with greater than 50% missing values, 221 features remained for modelling.

2.2 Why F2 Score, Not Accuracy

A model that predicts every transaction as legitimate scores 96.5% accuracy while catching zero fraud. Accuracy is therefore a useless metric for this problem. The chosen primary metric is the **F2 score** (F-beta with beta=2), which weights recall twice as heavily as precision. This reflects the cost asymmetry: a missed fraud (false negative) incurs the full transaction value loss plus chargeback fees, while a false alarm (false positive) causes customer friction that is recoverable. **PR-AUC** (average precision) is used for threshold-independent model comparison and architecture selection.

2.3 Imbalance Handling

Four strategies were evaluated on a 50,000-row stratified subsample using Logistic Regression as the base estimator (to isolate the effect of the resampling strategy from model power): SMOTE, ADASYN, class_weight='balanced', and threshold tuning on a default model. Threshold tuning achieved the best PR-AUC (0.322) on the subsample, while class_weight achieved the best F2 (0.377). For XGBoost and LightGBM, scale_pos_weight and is_unbalance were used respectively — no synthetic resampling was needed as tree models handle imbalance through the loss function directly.

2.4 Model Training

Three models were trained on 80% of the data (472,432 transactions) and evaluated on a held-out 20% validation set (118,108 transactions). Logistic Regression (SAGA solver, class_weight='balanced') served as the baseline. XGBoost and LightGBM were each tuned using Optuna Tree-structured Parzen Estimator (TPE) over 50 trials with a 10-minute timeout, optimising PR-AUC on the validation set. The final models were retrained on the full training set using the best hyperparameters found.

3. Results

3.1 Model Comparison

Model	PR-AUC	ROC-AUC	F2 Score	Recall	Precision	Threshold
Logistic Regression	0.3062	0.8372	0.3958	0.5437	0.1896	0.6633
XGBoost	**0.8242**	**0.9670**	**0.7669**	**0.7866**	**0.6970**	**0.4456**
LightGBM	0.7861	0.9622	0.7364	0.7687	0.6305	0.6435

Table 1: Validation set metrics for all three models. XGBoost (highlighted) achieves the best PR-AUC and F2 score. All thresholds are F2-optimised on the validation set.

XGBoost outperforms LightGBM by 3.8 percentage points in PR-AUC (0.824 vs 0.786) and surpasses the Logistic Regression baseline by 169%. The ROC-AUC of 0.967 indicates excellent overall discrimination, though PR-AUC is the operationally relevant metric given the severe class imbalance.

3.2 Precision-Recall Curves

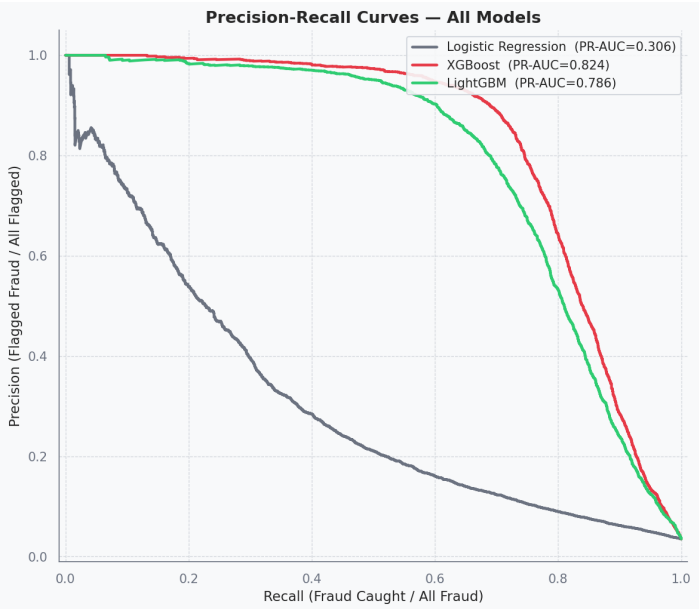

Figure 1: Precision-Recall curves for all three models. Higher area under the curve indicates better fraud detection at all operating thresholds. XGBoost dominates across the full range.

3.3 Confusion Matrix at Operating Threshold

At the F2-optimal threshold of 0.446, XGBoost correctly identifies 3,251 fraud transactions (true positives) while missing 882 (false negatives). It generates 1,413 false alarms on legitimate transactions.

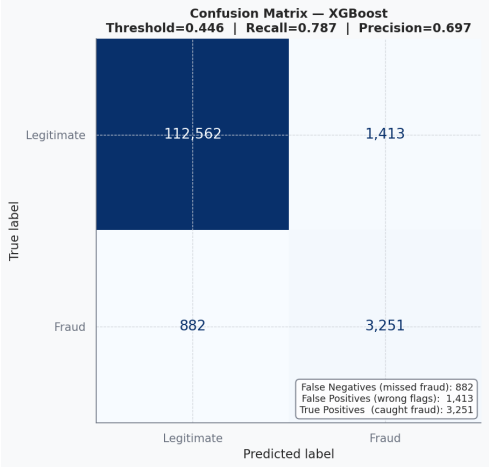

Figure 2: Confusion matrix at F2-optimal threshold (0.446). True Positives = caught fraud. False Negatives = missed fraud (highest cost).

3.4 Threshold Analysis

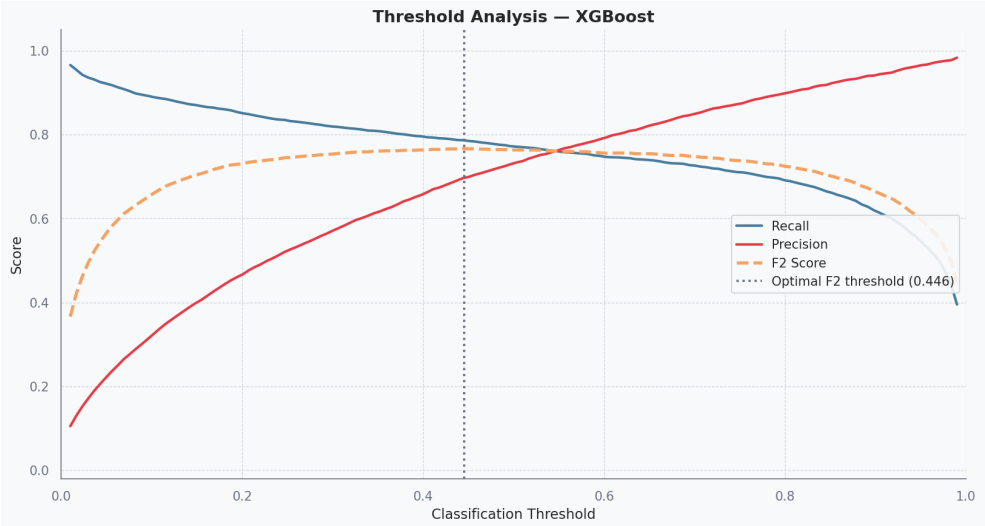

Figure 3: Precision, Recall, and F2 Score vs classification threshold. The vertical line marks the F2-optimal operating point. As threshold rises, precision increases and recall falls.

4. Model Card

A model card is a standardised summary of model characteristics, intended use, performance, and limitations. It enables downstream stakeholders to make informed decisions about deployment.

Model Name
XGBoost Fraud Classifier v1.0

Model Type
Gradient Boosted Decision Trees (XGBoost 3.x)

Intended Use
Binary classification of payment transactions as fraudulent or legitimate. Intended for use in a human-in-the-loop review queue and automated hard-block pipeline within a payments fraud prevention system.

Out-of-Scope Use
Not suitable for: credit scoring, identity verification, real-time streaming inference without re-evaluation of latency, or application to transaction data from domains substantially different from Vesta's payment network.

Training Data
IEEE-CIS Fraud Detection dataset. 472,432 training transactions. Date range: December 2017 to April 2018. Source: Vesta Corporation payment network.

Validation Data
118,108 held-out transactions (20% stratified split). Date range: April to May 2018.

Input Features
221 features: 212 numeric (transaction amount, Vesta V-features, card metadata, temporal features) + 9 categorical (ProductCD, card brand/type, email domain, M-fields).

Performance
PR-AUC: 0.8242 | ROC-AUC: 0.9670 | F2: 0.7669 | Recall: 0.7866 | Precision: 0.6970

Operating Threshold
F2-optimal: 0.4456 (review queue) | High-precision: 0.9900 (hard block)

Explainability
SHAP TreeExplainer provides global feature importance and per-transaction explanations. LIME provides local linear approximations for audit of individual decisions.

Training Date
23 April 2026

Retraining Cadence
Recommended quarterly, or when monthly PR-AUC on live traffic drops more than 3 percentage points below validation baseline.

5. Explainability

All model decisions are interpretable via SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). This section presents three XAI artefacts: global feature importance, a single flagged fraud explained, and a false positive dissected.

5.1 Global Feature Importance — SHAP Beeswarm

The beeswarm plot shows the top 20 features by mean absolute SHAP value across 1,000 sampled validation transactions. Each dot represents one transaction. Colour encodes feature value (red = high, blue = low). Features at the top exert the strongest influence on predictions. A positive SHAP value pushes the prediction toward fraud; negative pushes toward legitimate.

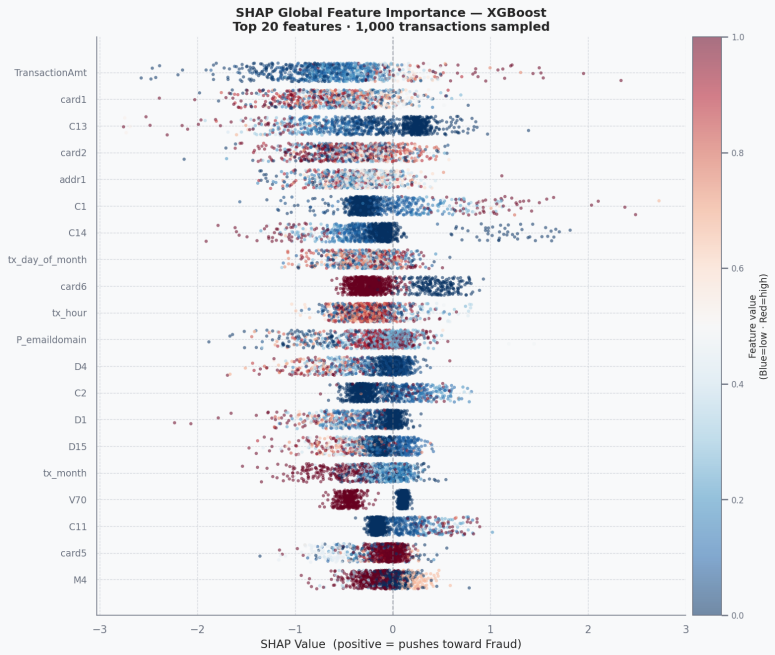

Figure 4: SHAP beeswarm plot — global feature importance. V-features (Vesta behavioural signals) dominate, confirming that transaction velocity and behavioural fingerprints are the primary fraud signals, not the transaction amount alone.

5.2 Why We Flagged This Transaction — SHAP Waterfall

The waterfall plot decomposes a single fraud prediction into the contribution of each feature. Starting from the model's expected output (base rate), each bar shows how one feature shifts the prediction up (toward fraud, red) or down (toward legitimate, blue). The final prediction probability is the sum of the base value and all feature contributions. This transaction was flagged with 100% confidence.

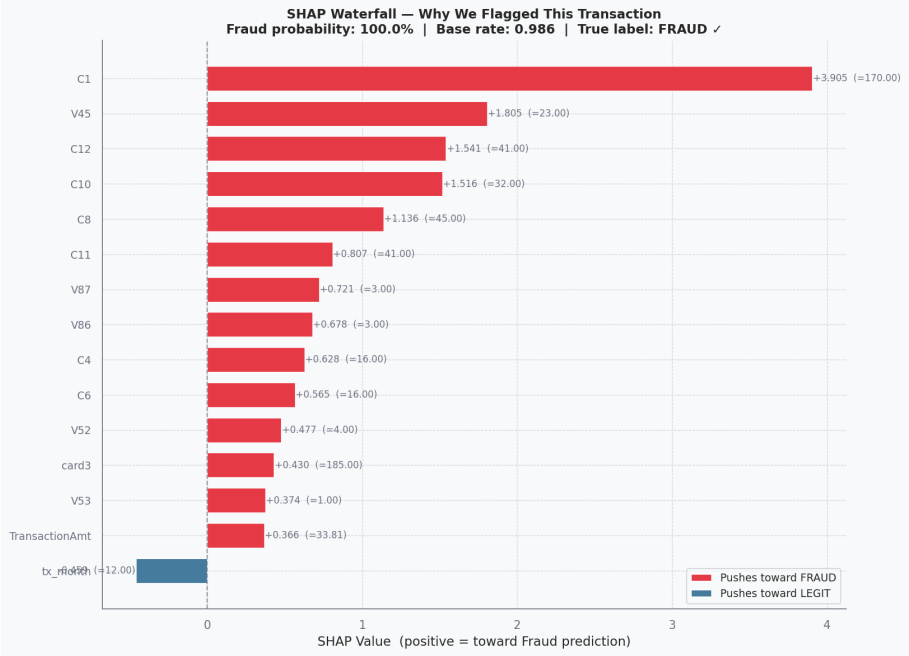

Figure 5: SHAP waterfall for the highest-confidence fraud transaction. Multiple V-features compound upward, while the transaction amount contributes a smaller directional push. This explanation would be shown to an analyst reviewing the flag.

5.3 Why We Incorrectly Flagged This Customer — LIME

The LIME plot explains a false positive: a legitimate transaction that the model flagged as fraud. LIME fits a local linear model around this specific transaction, identifying which feature values caused the incorrect classification. Understanding false positives is operationally important: it reveals edge cases where the model's learned patterns generalise poorly, and informs rule-based overrides or feature engineering to reduce false alarm rates.

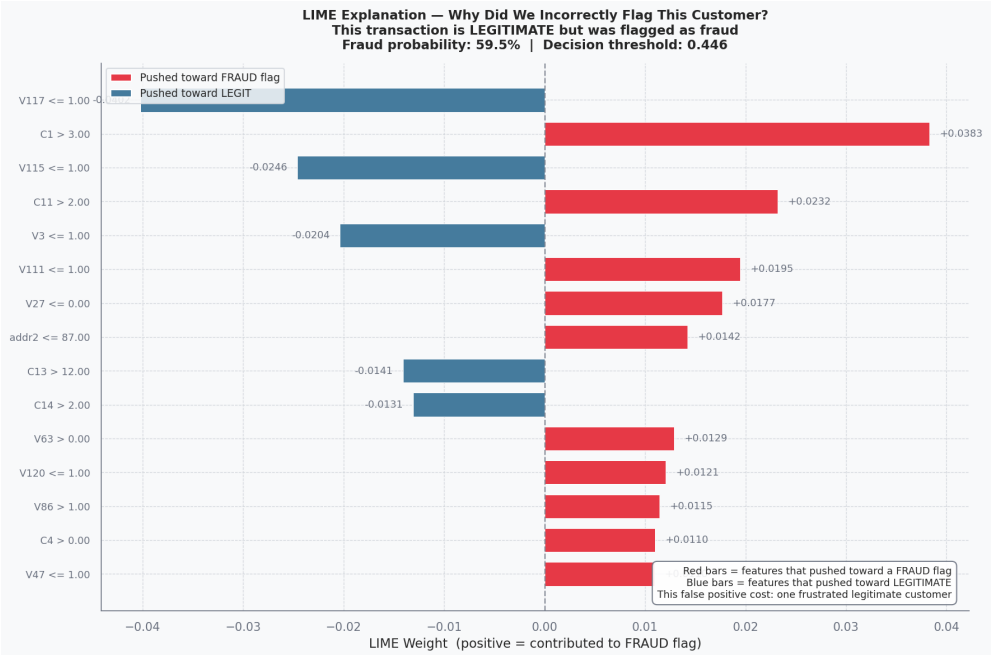

Figure 6: LIME explanation for a false positive (legitimate transaction incorrectly flagged as fraud). Red bars pushed the model toward a fraud decision; blue bars pushed back. The transaction cleared the decision threshold despite being legitimate, costing one customer friction event.

6. Limitations & Recommendations

6.1 Known Limitations

• **Temporal validity:** The model was trained on data from late 2017 to mid-2018. Fraud tactics evolve rapidly. Model performance should be monitored continuously and retraining triggered when PR-AUC degrades.

• **Feature opacity:** The Vesta V-features (V1-V321) are proprietary engineered signals whose exact definitions are not publicly documented. This limits full interpretability and makes feature engineering by the deployment team difficult without Vesta's documentation.

• **Identity coverage:** Only 24.4% of transactions have identity signals (device, browser). The model cannot leverage identity features for the majority of transactions, likely leaving recall on the table.

• **No real-time velocity features:** Features such as 'number of transactions on this card in the last 5 minutes' are not available in the static dataset. These are among the strongest fraud signals in production systems.

• **Subgroup fairness not evaluated:** Model performance was not disaggregated by geography, card issuer, or demographic proxies. Disparate false positive rates across customer segments would be a material concern for deployment.

• **Threshold sensitivity:** The F2-optimal threshold (0.446) was selected on the validation set and may not generalise perfectly to live traffic with different fraud rate or transaction mix.

6.2 Recommendations

• **Immediate deployment:** XGBoost at F2-threshold 0.446 for the analyst review queue. Hard-block threshold 0.990 for automatic decline.

• **Monitoring:** Track weekly PR-AUC, recall, and false positive rate on live traffic. Alert if PR-AUC drops below 0.78 (5 pp below validation).

• **Feature engineering:** Add real-time velocity features (card velocity in 1h, 24h windows) and graph-based network features connecting cards, emails, and devices. Expected 5-10 pp PR-AUC improvement.

• **Retraining cadence:** Quarterly full retraining on rolling 12-month window. Trigger immediate retraining if fraud rate shifts by more than 1 percentage point.

• **Fairness audit:** Before production deployment, conduct subgroup analysis by card type, product code, and transaction amount bucket. Ensure false positive rate is equitable across segments.

• **Online learning:** Consider a lightweight online gradient boosting layer trained on recent confirmed fraud labels to adapt to emerging fraud patterns without full retraining.

This report was generated programmatically from model artefacts and evaluation outputs. All figures and metrics reflect performance on the held-out validation set unless otherwise stated. Code and full reproducibility instructions are available in the project repository.